

# TLAMATINI

Complete Project Dossier: what the system does, how it works, how to use Tlamatini, complete tracked file tree, and effective line inventory



| Measure                   | Value                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| Generated                 | May 31, 2026 09:37 PM UTC-0600                                                                          |
| Current HEAD              | 1df130b - Adding ES32er Agent programmer, debugger and uploader, by angelahack1, to push version 1.12.0 |
| Resolved version          | 1.12.0 (git)                                                                                            |
| Tracked files             | 519                                                                                                     |
| Workflow agents           | 69                                                                                                      |
| Multi-Turn tools          | 76                                                                                                      |
| Skills                    | 26                                                                                                      |
| Total effective lines     | 116,777                                                                                                 |
| Total physical text lines | 159,672                                                                                                 |

# 1. What Tlamatini Is

- Tlamatini is a local-first AI developer assistant built with Django, Django Channels, LangChain, LangGraph, FAISS/BM25 retrieval, and a large in-repository agent application.
- She combines a browser chat surface, a Retrieval-Augmented Generation stack, a Multi-Turn tool executor, MCP-backed context providers, wrapped chat-agent runtimes, and a visual Agentic Control Panel for workflow design.
- She is designed for development operations: codebase analysis, file and directory context, command execution, Python execution, screenshots, web/search helpers, notifications and attention routing, DevOps tools, local model operation, Windows packaging and uninstall registration, first-person self-knowledge about her own runtime, and embedded-firmware control for both STM32F4 and ESP32-class boards.

## What the system does

- Answers codebase questions with loaded file or directory context.
- Uses hybrid retrieval to extract metadata, split content, rank source chunks, and respect context budgets.
- Gives operators GUI-first database maintenance through the new DB dropdown for backup and staged database replacement.
- Can pause before every state-changing Multi-Turn execution and ask the operator to approve or deny that exact step through the Ask Execs checkbox.
- Can raise a Windows attention signal when the browser needs the operator: Ask Execs prompts and Notifier events can flash Tlamatini's own taskbar presence and leave an uppercase banner in ``tlamatini.log``.
- Registers packaged installs in Windows ``Installed apps`` / ``Programs and Features`` with a real uninstall entry, so the release behaves like a normal installed application instead of only a shortcut bundle.
- Warns GPU-host operators before a directory-context load is likely to saturate VRAM and degrade embedding throughput.
- Exposes a coherent versioning surface across builds, runtime UI, logs, and an open health-check endpoint.
- Carries a first-person self-knowledge map so she can answer more accurately about her own architecture, ports, runtime modes, pages, and capabilities.
- Can command Kali Linux offensive-security tooling through MCP-Kali-Server for authorized recon, enumeration, web scanning, and assessment workflows.
- Can scaffold, author, build, flash, reset, and observe STM32F4 firmware through STM32er and the STM32 Template Project MCP, with a fail-safe preflight before any hardware mutation.
- Can scaffold, author, build, upload, and monitor ESP32-class firmware through ESP32er and PlatformIO Core, with zero-config bootstrap and a serial-aware preflight before hardware mutation.
- Can manage real desktop windows by title: focus them, tile them, resize them, list them, and close them deterministically through Win32 calls.
- Can drive a real Playwright browser through scripted interactive steps for logins, forms, assertions, downloads, extraction, and end-to-end UI checks.
- Can drive a live Unreal Engine 5 editor through the Unreal MCP plugin, from either Multi-Turn chat or the visual workflow canvas.
- Actively reaps orphaned Windows console-host and pool-child processes so long Multi-Turn or ACPX sessions do not leave misleading Tlamatini-icon ghosts in Task Manager.
- Runs checked Multi-Turn requests through request-scoped planning, capability selection, tool calls, observations, monitoring, and final synthesis.
- Can optionally inspect and modify her own bundled source tree in self-modify builds after verifying that ``TlamatiniSourceCode/`` is actually present.
- Launches wrapped copies of selected workflow agents in isolated runtime folders without mutating templates.
- Lets users design, validate, save, pause, resume, and stop visual workflows through the Agentic Control Panel.
- Turns successful Multi-Turn tool executions into starter ``flw`` workflows that can be inspected and validated in ACP.

- Packages the project into a distributable Windows release with installer and uninstaller tooling.

## How it works

- Browser UI sends chat and workflow requests through Django views and Channels WebSockets.
- RAG chains load selected file/directory context, retrieve relevant chunks, and build answer prompts.
- DB-menu actions validate directories or SQLite files in the browser, then call Django views that either copy the live database out or stage a replacement into `DB/ToLoad/db.sqlite3`.
- When Ask Execs is enabled, the synchronous Multi-Turn executor stops before each state-changing tool call, emits an `exec\_permission\_request`, and waits on `ExecPermissionBroker` until the browser sends Proceed or Deny.
- When the browser surfaces an Ask Execs prompt or a Notifier event, JavaScript can POST to `/agent/flash\_window/`; the backend then best-effort flashes the `Tlmatini.exe` console/taskbar window through `window\_flash.py` and prints an uppercase attention banner for the log.
- Installer-time registration writes a per-user HKCU Add/Remove Programs entry pointing at `Uninstaller.exe`, and frozen startup re-checks that entry through `windows\_app\_registration.self\_heal\_for\_frozen()` so older installs retroactively appear in Windows' uninstall surfaces.
- Before a heavy directory embedding run on supported NVIDIA hosts, a fail-open pre-flight guard can estimate VRAM pressure and surface a non-blocking warning in chat.
- Version resolution now flows through git tags, a runtime resolver module, generated build artefacts, and an open `/agent/version/` endpoint.
- A first-person self-knowledge file (`Tlmatini.md`) is injected into prompt construction for all chains, but loaded user context still outranks that self-reference when the request is a generic summary of the provided project.
- When Multi-Turn is enabled, the global planner selects context and tool stages before the executor binds only the relevant tools, including wrapped deterministic agents such as De-Compressor.
- Optional self-modify builds bundle `TlmatiniSourceCode/`; prompt rules require her to verify that directory exists before claiming she can inspect or change her own code.
- The Kalier path talks directly to the MCP-Kali-Server Flask API over HTTP with Python-stdlib `urllib`, auto-seeding the default box from `kali\_server\_url` in Config -> URLs before any one-off per-call override is applied, and captures one atomic `INI\_SECTION\_KALIER` block per run.
- The STM32er path spawns the STM32 Template Project MCP stdio server, performs the MCP initialize handshake, runs exactly one requested tool or composite action, and emits one atomic `INI\_SECTION\_STM32ER` block with the result, project directory, and stage metadata.
- Before any flash-capable STM32er action, a critical-mission preflight validates the arm-none-eabi toolchain, STM32CubeIDE, programmer path, ST-LINK presence, and STM32F-family match; compile-only steps can run boardless, but unsafe hardware mutations are refused fail-safe.
- The ESP32er path resolves or bootstraps PlatformIO Core, invokes `pio` subcommands directly with Python-stdlib process control, validates project and serial-port readiness, and emits one atomic `INI\_SECTION\_ESP32ER` block with stage, project, port, and stdout/stderr payloads.
- The Windower path uses Win32 APIs plus the cross-process `AttachThreadInput` focus-transfer dance to locate windows by title and apply one lifecycle action while still returning structured geometry/state fields.
- The Playwrighter path loads a declarative step list, drives Playwright against Chromium/Firefox/WebKit, and emits one atomic `INI\_SECTION\_PLAYWRIGHTER` block with status, assertions, extracted values, and the final URL.
- The Unreal path opens a TCP socket to the Unreal MCP plugin, sends one `{"type": "command", "params": {...}}` payload, captures the JSON reply, and emits one `INI\_SECTION\_UNREALER` block for downstream logic.
- After spawn-capable tool calls and again after the final answer, the orphan reaper can sweep dead descendants, orphaned `conhost.exe` companions, and stale pool-linked processes without ever raising into the chat path.
- Tool calls execute in the backend, append observations, and may create wrapped runtime copies under `agent/agents/pools/\_chat\_runs/`.
- On the next full start-up, `manage.py` can swap a staged database into place before Django imports, while archiving the previous live database under `DB/Older/<timestamp>/`.

- ACP flows deploy session-scoped pool instances, wire config values, validate NxN graph rules, and execute through Starter-driven flow semantics.
- Build scripts collect static assets, bundle Django/Python resources, add agent templates, and assemble `pkg.zip`, `Uninstaller.exe`, and `dist/Tlamatini\_Release/`.

## 2. Architecture Layers

| Layer              | Role                                                                                                                       |
|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| Browser interfaces | Chat page plus Agentic Control Panel templates and JavaScript modules.                                                     |
| Django/Channels    | Authentication, views, WebSockets, session state, message persistence, and ASGI startup.                                   |
| RAG and context    | Metadata extraction, text splitting, FAISS/BM25 retrieval, context budgeting, and fallback behavior.                       |
| Multi-Turn engine  | Capability registry, global execution planner, explicit tool loop, answer parsing, and answer-success classification.      |
| Tools and agents   | Core tools, MCP context providers, wrapped chat-agent launchers, and the current visual workflow agent templates.          |
| Packaging          | PyInstaller build scripts, shortcut registration, `.flw` association, installer, uninstaller, and release folder assembly. |

### Design principles

- Evidence-first answers: Tlamatini grounds responses in selected project context and hybrid retrieval rather than freeform model memory.
- Explicit orchestration: checked Multi-Turn uses a visible tool loop, capability scoring, and staged planning instead of a single opaque call.
- Operational reversibility: risky changes such as database replacement are staged, archived, and delayed to the only safe window instead of hot-swapped mid-session.
- Fail-open diagnostics: GPU pressure probes and session-restore safeguards warn early without breaking CPU-only or degraded environments.
- Runtime isolation: wrapped chat-agent copies run in session-scoped folders so template agents remain pristine while live runs stay inspectable.
- Self-knowledge with scope discipline: she can talk accurately about herself without letting self-reference override a user-loaded project context.
- Operator truth over vibes: Exec Report tables, tlamatini.log, skill audits, and ACPX transcripts make the system auditable after execution.

## 3. Installation, Configuration, and Everyday Use

### Installation essentials

- Python 3.12.10 is the strongly recommended source-mode version in the README, and the codebase has been tested most deeply there.
- Source installs require a clone, virtual environment, dependency install from `requirements.txt`, migrations, a superuser, static collection, and then the web server.
- You can run either the checked-in cloud/back-end defaults from `Tlmatini/agent/config.json` or a local Ollama-backed configuration with matching model names.
- Packaged Windows installs create a default `user / changeme` account; manual source installs use your own `createsuperuser` account instead.

### Configuration essentials

- Source mode resolves `Tlmatini/agent/config.json`; frozen builds resolve `config.json` next to the executable; `CONFIG\_PATH` overrides both.
- Core keys include `embedding-model`, `chained-model`, `ollama\_base\_url`, `ollama\_token`, `enable\_unified\_agent`, `unified\_agent\_model`, and `unified\_agent\_max\_iterations`.
- The checked-in default model baseline moved again in the recent Git window: the shared config now favors `kimi-k2.6:cloud`, so source or frozen installs that keep the shipped config should be documented as cloud-first unless the operator intentionally swaps models.
- URL configuration now also includes `kali\_server\_url`, the STM32er bootstrap fields `stm32\_mcp\_server\_script`, `stm32\_mcp\_python`, `stm32\_template\_dir`, `stm32\_ide\_root`, `stm32\_mcp\_repo\_url`, and `stm32\_mcp\_install\_dir`, plus ESP32er's `pio\_executable` and `pio\_core\_dir`, all edited from `Config -> URLs` and inherited automatically by the chat-side wrapped tools.
- The chat-side Config -> Models and Config -> URLs dialogs are now first-class configuration surfaces, and they can explicitly ask the operator to reconnect when saved values change live-session assumptions.
- The separate DB dropdown is not a config editor: it is a maintenance surface for copying the live SQLite database out or staging a replacement for the next full start-up.
- Multi-Turn is toggled from the chat toolbar, but it depends on the unified-agent configuration and the selected model/base-url pairing being valid; the current default iteration ceiling is 4096, and Ask Execs only becomes available when Multi-Turn itself is on.
- Image interpretation can run through Claude-backed cloud paths or Qwen/Ollama-backed local paths, and remote Ollama can be protected with a bearer token.

### DB menu and database swap-in

- The new DB dropdown gives operators two GUI-first maintenance paths: `Backup database` copies the live SQLite file out, and `Set DB` stages a chosen `db.sqlite3` for the next full start-up.
- Both dialogs are live-validated in the browser and now expose Browse buttons that open native host-side pickers for folders or `db.sqlite3` files.
- Backup is read-only and uses the currently live database path; Set DB never hot-swaps the file mid-session because Django already holds SQLite open.
- Set DB writes the selected file to `DB/ToLoad/db.sqlite3`; the real swap happens only at the top of `manage.py` before Django imports anything.
- When that swap runs, the previous live database is moved into `DB/Older/<timestamp>/db.sqlite3`, creating a built-in rollback trail instead of overwriting history.
- Reconnect is not enough for this path: the operator must fully restart Tlmatini so the pre-Django swap window opens again.
- If the staged file is bad or locked, startup fails open: Tlmatini logs the error and continues with the previous live database.

## Versioning system

- Tlamatini now follows Semantic Versioning 2.0.0 with git tags as the single source of truth: you tag, then you build, instead of hand-editing version strings across files.
- The build path resolves a version once and propagates it into generated runtime metadata, Win32 VERSIONINFO resources, and the release-folder naming convention.
- On untagged commits the resolver falls back honestly to a git-derived development version instead of failing the build or lying about the release state.

## Version surfaces

- Operators can now see the running version in the About dialog, the startup banner, the open ``GET /agent/version/`` health-check endpoint, and Windows file properties on the built executables.
- Packaged installs also project the same release identity into Windows' Installed-apps metadata through the ARP ``DisplayVersion`` field, so the uninstall surface and the binaries agree on the version being removed.
- The runtime resolver lives in ``Tlamatini/agent/version.py``, while the build-oriented coordination logic lives in the repo-root ``versioning.py`` and the longer policy notes live in ``VERSIONING.md``.
- Build overrides are supported in a predictable order: CLI ``--version``, then ``TLAMATINI_VERSION``, then git describe, then the sentinel ``0.0.0+unknown``.

## Self-knowledge in v1.8.0

- Version ``1.8.0`` gives Tlamatini a first-person self-knowledge map in ``Tlamatini/agent/Tlamatini.md``, so she can answer more accurately about her own architecture, runtime modes, open ports, pages, and internal capability surface.
- That self-reference is injected into all prompt chains through ``prompt.pmt`` and ``agent/rag/config.py``, but it fails open if the file is missing or unreadable and it never overrides a user-loaded project when the request is a generic summary of the provided context.
- The language contract matters too: when a pronoun is used for Tlamatini in the documentation or prompt guidance, she is referred to as ``she`` / ``her``, matching the updated handbook identity rules.

## Self-modify builds

- Self-modification is a separate capability axis from frozen-vs-source runtime: only builds created with ``python build.py --self-modify`` bundle ``TlamatiniSourceCode/`` next to the application.
- When that directory is present, she can inspect and modify her own shipped source tree; when it is absent, she must say so plainly and fall back to her injected self-knowledge plus the surrounding docs.
- The build pipeline now announces that choice explicitly, so operators can tell whether a release is self-modify-capable before asking her to work on herself.

## Multi-Turn 4096-turn autonomy

- The unified-agent loop now defaults to 4096 iterations instead of 256, giving long autonomous operator runs much more room before they exhaust the turn budget.
- That expansion is about conversational/tool-loop depth, not about blindly firing more tools at once: the planner's selected-tool cap still keeps the tool surface bounded per request.
- Operationally, the bigger ceiling helps long workflows, while duplicate-call guards and the dedicated ``chat_agent_sleeper`` tool remain the antidote to accidental busy-polling loops.

## Ask Execs in Multi-Turn

- Introduced in ``v1.10.0`` and still part of the current ``v1.12.0`` surface, ``Ask Execs`` is the Multi-Turn-only safety modifier that makes Tlamatini ask before each state-changing Tool, MCP, wrapped agent, or skill-backed execution instead of running it immediately.
- The permission dialog is explicit and auditable: it names the Tool or Agent family, the underlying raw tool name, the full parameters, the program or command to be executed, and the shell or execution surface involved.

- Proceed runs that one step and then prompts again at the next state-changing step; Deny halts the entire chain immediately and appends a red `Execution interrupted` banner even when Exec Report itself is off.

## Ask Execs runtime path

- Under the hood, `agent/exec\_permission.py` provides `ExecPermissionBroker`, which lets the synchronous worker-thread executor emit a permission request onto the WebSocket event loop and then block on a `threading.Event` until the browser replies.
- The gate sits after deduplication and quota checks, so skipped calls never prompt and denied calls never appear as executed rows; only work that truly ran lands in Exec Report.
- The round-trip is fail-safe: browser disconnect, emit failure, cancel, or broker shutdown all resolve to Deny, so an unconfirmed state-changing action never slips through just because the UI vanished at the wrong time.

## Windows attention issuing

- The current Windows attention path is explicit and local: the browser calls `POST /agent/flash\_window/`, and the backend routes that request through `agent/window\_flash.py`.
- That helper can flash the `Tlmatini.exe` console/taskbar window with `FlashWindowEx` and always prints an uppercase attention banner, so the signal survives in `tlmatini.log` even when the browser is minimized.
- Two concrete reasons are wired today: Ask Execs execution approval prompts and Notifier notifications. The path is best-effort and fail-safe, degrading cleanly on non-Windows or windowless launches.

## Windows Installed-apps registration

- Introduced in `v1.11.0` and still carried by the current `v1.12.0` release, the frozen install now behaves like a real Windows application: `install.py` writes a per-user HKCU Add/Remove Programs entry so Tlmatini appears in Settings -> Apps -> Installed apps and in the legacy Programs and Features list.
- The entry carries `DisplayName`, `DisplayVersion`, `InstallLocation`, `DisplayIcon`, `UninstallString`, `QuietUninstallString`, `NoModify`, `NoRepair`, and best-effort `EstimatedSize`, all pointing at the bundled `Uninstaller.exe` without requiring administrator rights.
- The matching runtime self-heal in `agent/apps.py` calls `windows\_app\_registration.self\_heal\_for\_frozen()` on every frozen launch, so installs created before this feature existed can appear in Windows' uninstall UI after the next normal app start.

## De-Compressor agent

- De-Compressor is the deterministic archive worker for compression and decompression tasks, deciding direction from whichever side exposes a recognized archive extension.
- Supported archive families are `.gz`, `.zip`, `.7z`, `.tar.gz`, and `.gz.tar`; file-to-folder extraction and file-or-directory packing are both documented in the README and Book.
- Password handling is explicit: `passwordless=true` skips it, while `passwordless=false` requires the `DE\_COMPRESSER\_PWD` environment variable and fails fast if the secret is missing.

## De-Compressor integration and fallback behavior

- The agent is reachable from both operator surfaces: ACP canvas nodes wire through dedicated connection-update views, and checked Multi-Turn can invoke it through `chat\_agent\_de\_compressor`.
- Format engines are practical rather than magical: stdlib `gzip` and `zipfile` cover core cases, `7z` is preferred for encrypted `7z` or `zip`, and `py7zr` was added to `requirements.txt` as the Python fallback.
- Every run emits an `INI\_SECTION\_DE\_COMPRESSER<<< ... >>>END\_SECTION\_DE\_COMPRESSER` block and still triggers `target\_agents`, so downstream Parametrizer or Raiser logic can branch on `success=true|false` instead of guessing from prose.

## Unreal MCP and the Unreal agent

- Unreal MCP is a UE5 plugin that runs inside the editor and listens on `127.0.0.1:55557` for one JSON command per TCP connection; Tlmatini is the client side of that link, not the plugin host.



- The Unreal workflow agent forwards any command the connected plugin build exposes, up to a 53-command surface across nine categories: actor manipulation (incl. viewport screenshots), Blueprint authoring and node wiring, input mappings, UMG widget building, in-editor Python/console execution, level I/O, asset import, and material authoring. Headless build/cook/test is out of scope (it needs UnrealEditor-Cmd, not the editor socket).
- The same integration is available in both operator surfaces: checked Multi-Turn via ``chat_agent_unrealer``, and ACP canvas flows via the visual Unreal node plus Parametrizer chaining.

## Extended Unreal MCP surface

- Unreal's documentation now points at the public ``XAIHT/XaihtUnrealEngineMCP`` fork, the Unreal Engine MCP variant developed specifically for Tlamatini.
- That fork exposes the extended 53-command, nine-category surface documented in README and Book: not only the base editor/Blueprint/UMG verbs, but also system, level, asset, material, screenshot, viewport, and in-editor Python paths.
- The practical message for operators is simple: Unreal forwards the connected plugin's verb surface directly, so Tlamatini's client does not need a new release every time the plugin adds another supported command.

## Installing the UE5 plugin and smoke-testing it

- Install the upstream plugin into ``<YourProject>/Plugins/UnrealMCP/``, enable it from ``Edit -> Plugins``, restart UE5, and confirm the Output Log says ``UnrealMCP listening on 127.0.0.1:55557``.
- Tlamatini does not compile or embed the plugin; the Unreal editor must already be running with the listener bound before any Unreal call can succeed.
- A seeded smoke test ships in the Prompts table: ``Unreal MCP End-to-End Editor Drive`` walks through sanity-check, actor spawn, Blueprint creation/compile, and UMG widget assembly.

## Orphan-process cleanup

- Tlamatini now ships a three-tier orphan reaper in ``Tlamatini/agent/orphan_reaper.py`` focused on Windows console-host leftovers such as ``conhost.exe`` and ``openconsole.exe``.
- Tier 1 runs after spawn-capable Multi-Turn tool calls, Tier 2 runs once after the final answer in a background thread, and Tier 3 runs again during application shutdown through the same cleanup path that already tears down pools.
- The candidate set stays narrow on purpose: dead descendants of the current process tree, orphaned console hosts tied to that tree, and pool-linked processes whose ``cmdline`` still points into ``agents/pools/...`` even though tracking records are gone.

## Orphan-process prevention and survivor reporting

- Prevention landed alongside cleanup: Windows spawn sites now use ``CREATE_NO_WINDOW | DETACHED_PROCESS | CREATE_NEW_PROCESS_GROUP`` with stdio piped to ``DEVNULL``, so the console host is usually never allocated in the first place.
- The ACPX runtime now kills full process trees instead of only top-level wrappers, and pool-agent scripts gained a conservative ``subprocess.Popen`` guard so forgotten descendants still inherit ``CREATE_NO_WINDOW``.
- If something survives both cleanup tiers, Tlamatini sends a second chat bubble listing each surviving ``name + PID``; the reaper never raises into the user path because a cleanup crash would be worse than the leftovers it tried to remove.

## How to use it

- Run from source: create a virtual environment, install requirements, migrate, create a superuser, collect static files, and start Django.
- Open ``/agent/`` for chat. Load a file or directory context before asking codebase-specific questions.
- Keep Multi-Turn unchecked for direct Q&A; enable Multi-Turn for tasks that need tools, wrapped agents, monitoring, or workflow seeding.
- Tick ``Ask Execs`` when you want human approval before each state-changing Multi-Turn step; it is disabled until Multi-Turn is on, and a single Deny stops the whole chain with an explicit red interruption banner.

- When you are using a packaged install on Windows 10 or Windows 11, uninstall it through Settings -> Apps -> Installed apps or the legacy Programs and Features entry, not by manually deleting the folder.
- If an Ask Execs approval dialog or a Notifier event needs you while the browser is buried, watch for Tlamatini's taskbar-attention flash and the matching uppercase banner in `tlamatini.log`.
- If you want her to inspect or modify herself, verify that `TlamatiniSourceCode/` exists in the current build first; self-modify is optional and absent builds must be treated honestly as read-only about their own code tree.
- For authorized Kali Linux assessments, run MCP-Kali-Server on the Kali box, set `Config -> URLs -> Kali server (Kali)` once, and then call `chat\_agent\_kalier` from Multi-Turn with the desired `action` and `target` without repeating the box URL each turn.
- For STM32 firmware work, install STM32CubeIDE, leave `Config -> URLs -> STM32 MCP server script` blank for zero-config bootstrap, and then call `chat\_agent\_stm32er` from Multi-Turn with one `action` at a time such as `validate`, `create\_project`, `write\_source`, `build`, `build\_and\_flash`, `serial\_session`, or `live\_monitor`.
- For ESP32 firmware work, leave `Config -> URLs -> pio\_executable` blank for zero-config PlatformIO bootstrap, then call `chat\_agent\_esp32er` from Multi-Turn with actions like `bootstrap`, `validate`, `create\_project`, `write\_source`, `build`, `upload`, `build\_and\_upload`, `monitor`, or `monitor\_session`.
- For desktop-window control, call `chat\_agent\_windower` from Multi-Turn to focus, tile, resize, list, or close a window by title, or model the same action in ACP with the Windower node.
- For interactive web automation, call `chat\_agent\_playwrighter` from Multi-Turn with a `steps\_json` script, or author the same step list visually with the Playwrighter node on the canvas.
- For Unreal Engine work, enable the Unreal MCP plugin inside a live UE5 project first, then call `chat\_agent\_unrealer` from Multi-Turn or use the visual Unreal node on the canvas.
- Archive jobs can now be described directly in Multi-Turn or modeled visually in ACP: De-Compressor infers compress vs decompress from the `input` or `output` extension.
- If a second post-answer warning bubble ever lists surviving `name + PID` entries, treat it as an honest cleanup report and end the listed processes manually from Task Manager if needed.
- Use the `ACPX-Skills` navbar menu when you need to browse, enable/disable, diagnose, or reload the shipped SKILL.md catalog without asking the LLM to be your admin surface.
- Use the DB dropdown when you need a safe database snapshot or want to stage a different `db.sqlite3` for the next start-up without hot-swapping the live SQLite file.
- Open `/agentic\_control\_panel/` to drag agents, connect them, configure each node, validate, start, pause/resume, stop, and save `.flw` workflows.
- Use `python build.py`, `python build\_uninstaller.py`, and `python build\_installer.py` only when producing a packaged Windows release.

## Agent descriptions and catalog source of truth

- The authoritative human-readable source for workflow-agent descriptions is `agents\_descriptions.md` at the repo root, not an embedded JavaScript map or a hard-coded Django list.
- Current validation compares live template directories, `agents\_descriptions.md` rows, and the README / Book bestiary counts so the generated dossier stays aligned with the running product.
- The ACP sidebar hover tooltip and the right-click Description dialog both resolve through that markdown file first; `README.md` is only a legacy fallback when the dedicated description file is absent.

## Reviewer and Analyzer

- The workflow catalog now includes two new high-value specialists: Reviewer and Analyzer, lifting the canvas inventory to 64 agents and the seed-skill catalog to 23 SKILL.md packages.
- Reviewer is LLM-powered: it resolves a git diff for `repo\_path`, reviews it with a senior-engineer prompt, and emits an `INI\_SECTION\_REVIEWER` block whose first routable field is `verdict = APPROVE | REQUEST\_CHANGES | COMMENT`.
- Analyzer is deterministic and scanner-driven: it runs whichever of `bandit`, `semgrep`, `ruff`, `eslint`, `gitleaks`, and `pip-audit` are installed on PATH over `target\_path`, then emits an `INI\_SECTION\_ANALYZER` block with `status` and

`total\_findings` for downstream routing.

- Both agents always trigger `target\_agents`, so a downstream Forker can branch on `{verdict}`, `{status}`, or `{total\_findings}` instead of scraping prose.
- They are intentionally canvas-only workflow agents: there is no wrapped `chat\_agent\_reviewer` or `chat\_agent\_analyzer`, and therefore no duplicate Exec Report row family for those names.
- The chat-side counterparts live in the ACPX skill catalog instead: `code-review` exposes senior-engineer git-diff review, while `security-audit` exposes the deterministic multi-scanner sweep.

## Operator surface counts

- The live operator surface now stands at 69 workflow agents, 76 Multi-Turn tools, 12 ACPX tools, and 26 skills.
- Source inspection confirms the newer total: 44 wrapped chat-agent tools in `chat\_agent\_registry.py`, which combines with 20 core Python tools and 12 ACPX/Skill tools for 76 Multi-Turn tools overall.
- The workflow-agent and wrapped-tool totals align cleanly with the handbooks, while the skill total is validated from the on-disk `agent/skills\_pkg` catalog so the dossier stays honest even when a simplified markdown summary lags behind the live tree.
- This matters operationally because the planner never binds everything at once: the documented default `max\_selected\_tools` cap stays at 20, so breadth of capability does not mean uncontrolled tool sprawl per turn.

## Kalier in v1.7.1

- Kalier remains the 67th workflow agent in `v1.7.1`, but its newest upgrade is usability: Tlmatini now acts as the embedded MCP-Kali-Server client for chat-side runs.
- It can issue one capability per run, including `nmap`, `gobuster`, `dirb`, `nikto`, `sqlmap`, `metasploit`, `hydra`, `john`, `wpscan`, `enum4linux`, arbitrary `command`, or a safe `health` probe of the remote server.
- The agent emits `INI\_SECTION\_KALIER` with `action`, `endpoint`, `subject`, `return\_code`, `success`, `timed\_out`, and `server\_url`, so downstream Forker or Parametrizer logic can branch on results without scraping prose.
- The practical operator result is simpler prompts: after the Kali box URL is configured once, normal Multi-Turn requests no longer repeat `server\_url` unless you intentionally override it for a one-off target.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool `chat\_agent\_kalier` now auto-injects the configured `kali\_server\_url`, while the visual Kalier canvas node still stores `server\_url` explicitly in YAML per node.
- Kalier talks straight to the Kali-side Flask API over HTTP using Python-stdlib `urllib`, so it stays self-contained in the pool subprocess and works the same in source or frozen builds; the embedded-client injection fails open if the config value is blank or unreadable.
- Authorized use only: the intended operator flow is an in-scope lab, CTF, or permitted engagement, often with `ssh -L 5000:localhost:5000 user@KALI\_IP` tunneling a remote Kali box back to `http://127.0.0.1:5000`, and the repo now includes `Tlmatini-Kali-Setup.md` for the zero-client walkthrough.

## STM32er in v1.9.0

- STM32er is the new 68th workflow agent in `v1.9.0`: Tlmatini's bridge to the `STM32 Template Project MCP`, built so she can scaffold, author, build, flash, reset, and observe STM32F4 firmware without driving the STM32CubeIDE GUI.
- The operator promise is zero-config bootstrap: leave `server\_script` blank and STM32er downloads or zip-falls-back to the MCP server on first use, installs `mcp` and `pyserial` if needed, validates, and caches the result so the user only installs STM32CubeIDE plus Tlmatini.
- Before any compile-or-hardware action, STM32er runs a critical-mission fail-safe preflight over the arm-none-eabi toolchain, STM32CubeIDE, programmer path, ST-LINK probe, and target family; compile-only steps can run boardless, but flash, erase, reset, serial, and SWD/live-memory operations are refused when the environment or device is wrong.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool `chat\_agent\_stm32er` takes one `action` per call, while the visual STM32er canvas node stores the same fields in YAML and triggers downstream agents on both success and failure.
- The tool surface includes the full project lifecycle plus hardware-in-the-loop composites: `validate`, `bootstrap`, `create\_project`, `write\_source`, `build`, `build\_and\_flash`, `serial\_session`, `live\_monitor`, and the rest of the 23 MCP

verbs, with every run emitting an ``INI_SECTION_STM32ER`` block for Forker or Parametrizer routing.

- Config -> URLs now seeds the chat path with ``stm32_mcp_server_script``, ``stm32_mcp_python``, ``stm32_template_dir``, ``stm32_ide_root``, ``stm32_mcp_repo_url``, and ``stm32_mcp_install_dir``, so firmware prompts normally describe only the task and target board instead of the plumbing.

## ESP32er in v1.12.0

- ESP32er is the new 69th workflow agent in ``v1.12.0``: Tlmatini's direct bridge to PlatformIO Core, built so she can scaffold, author, build, upload, and monitor ESP32-class firmware without relying on an external MCP server or IDE.
- The operator promise is zero-config bootstrap: leave ``pio_executable`` blank and ESP32er downloads or pip-falls-back to PlatformIO Core on first use, validates it, and caches it under a per-user directory so the user installs only the board USB driver plus Tlmatini.
- Before any build-or-upload action, ESP32er runs a serial-aware preflight over ``pio`` resolution, project shape, and connected ports; upload and monitor require a real serial device, while non-esp8266 targets are warned about rather than hard-refused because PlatformIO is intentionally multi-target.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_esp32er`` takes one ``action`` per call, while the visual ESP32er canvas node stores the same fields in YAML and triggers downstream agents on both success and failure.
- The tool surface covers environment/meta (``bootstrap``, ``validate``, ``system_info``, ``boards``), project lifecycle (``create_project``, ``write_source``, ``read_source``, ``list_sources``, ``clean``), build and flash (``build``, ``upload``, ``build_and_upload``, ``list_artifacts``), serial HIL (``device_list``, ``monitor``, ``monitor_session``), and package / QA paths (``pkg_install``, ``pkg_list``, ``pkg_update``, ``check``, ``test``), with every run emitting an ``INI_SECTION_ESP32ER`` block for Forker or Parametrizer routing.
- Config -> URLs now seeds the chat path with ``pio_executable`` and ``pio_core_dir``, so firmware prompts usually describe only the board, project, and task while the wrapped tool auto-injects the PlatformIO runtime plumbing.

## ESP32 Template Project reference baseline

- The new ``ESP32TemplateProject`` repository is the known-good baseline documented in BookOfTlmatini's bonus chapter: a plain PlatformIO project, not a server, meant to prove an ESP32 board and toolchain are healthy before larger firmware work.
- It mirrors ESP32er's grain: ``platformio.ini``, ``src/``, ``include/``, ``lib/``, ``test/``, a blinking ``main.cpp``, serial output at 115200, and GitHub-ready docs/CI so the reference project does not silently rot.
- Operationally, ESP32er can either point at a checkout of that template (``project_dir``) or scaffold an equivalent from scratch with ``action='create_project``, then carry the directory through build, upload, and monitor steps.

## Windower on Multi-Turn and canvas

- Windower is the new 66th workflow agent: a deterministic Win32 window manager that finds an application window by title and performs one lifecycle operation on the window itself instead of clicking inside it.
- It supports ``focus``, ``minimize``, ``maximize``, ``restore``, ``move``, ``resize``, ``move_resize``, ``close``, ``topmost``, ``untopmost``, ``arrange``, and ``list``, with ``substring``, ``exact``, or ``regex`` title matching plus ``match_index`` for duplicate titles.
- The agent emits ``INI_SECTION_WINDOWER`` with ``action``, ``window_title``, ``matched``, ``match_count``, ``state``, ``left``, ``top``, ``width``, and ``height``, so downstream Forker or Parametrizer logic can branch on presence, state, or geometry.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_windower`` accepts free-form key=value requests, while the visual Windower canvas node stores the same operation fields in YAML.
- Windower is the desktop-window sibling of Mouset and Keyboarder: use Windower when the goal is the window as a whole, Mouset for controls inside it, and Keyboarder for text entry into it.
- Because Windower changes real window state, its executions appear in Exec Report and it always triggers ``target_agents`` whether the action succeeds or fails.

## Playwrighter in v1.5.0

- Playwrighter is the new 65th workflow agent in ``v1.5.0``: a deterministic Playwright-powered browser automator for Chromium, Firefox, or WebKit that executes ordered interactive steps instead of static fetches.

- It covers the gap between Crawler and Googler: logins, multi-step forms, wizard clicks, JS-rendered SPA scraping, screenshots after interaction, assertions, downloads, and authenticated end-to-end UI checks.
- The agent emits `INI_SECTION_PLAYWRIGHTER`` with ``start_url``, ``final_url``, ``status``, ``steps_run``, ``assert_result``, and extracted values so downstream Forker or Parametrizer logic can branch on pass/fail or reuse scraped data.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_playwrighter`` takes the whole script as ``steps_json``, while the visual Playwrighter canvas node stores the same declarative step list in YAML.
- Session continuity is built in: ``headless: false`` lets operators watch it drive, and ``storage_state_in`` / ``storage_state_out`` carry login state across runs without forcing a manual browser setup each time.
- Because Playwrighter is state-changing, its executions appear in Exec Report and it always triggers ``target_agents`` whether the run succeeds or fails.

## Reviewer precision patch in v1.4.1

- The ``v1.4.1`` Reviewer refinement tightened accuracy rather than adding a new surface: when ``diff_ref`` is empty, the review prompt labels the diff as the uncommitted working tree plus staged area, so the model must not describe those findings as already committed or pushed.
- The same patch teaches the model Tlamatini's managed-secret convention: ``agent/config.json`` and selected ``agent/agents/*/config.yaml`` files can hold live local credentials in a keyed working copy, while ``regen_secrets.py --mode push-able`` scrubs them back to placeholders before commit.
- That guidance is mirrored into the ``code-review`` SKILL.md package too, keeping the chat-surface review behavior and the canvas Reviewer agent aligned on commit-state wording and secret-severity expectations.

## Native dialogs and Tkinter removal in v1.4.2

- The current tagged release ``v1.4.2`` removes Tkinter from the unstable runtime-facing dialog path and replaces it with ``Tlamatini/agent/native_dialogs.py``, a native Windows dialog bridge used by browser-triggered pickers.
- This change pairs with the existing DB and operator dialogs: file and folder selection still feels local and GUI-first, but the fragile Tkinter dependency is no longer part of the interactive runtime path that users trigger from chat or ACP surfaces.
- The patch arrived with dedicated tests (``test_native_dialogs.py``) and with follow-up orphan-reaper/runtime adjustments, so the release reads as a stability pass rather than a cosmetic refactor.

## ACPX-Skills menu

- The new ``ACPX-Skills`` navbar menu gives operators four direct actions over the skill catalog: Browse Skills, Configure Skills, Diagnostics, and Reload Registry.
- ``Configure Skills`` flips ``Skill.enabled`` exactly the way MCPs and Tools are toggled, so disabled skills disappear from ``list_skills`` and reject ``invoke_skill`` with ``SKILL_DISABLED`` instead of silently half-working.
- The diagnostics view cross-checks skill dependencies against disabled tools, disabled MCPs, missing ACPX agents, and orphan database rows whose SKILL.md disappeared from disk.

## Source-mode bootstrap commands

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python Tlamatini/manage.py migrate
python Tlamatini/manage.py createsuperuser
python Tlamatini/manage.py collectstatic --noinput
python Tlamatini/manage.py runserver --noreload
```

## 4. Ollama Setup Without Administrative Rights

- Open a normal PowerShell window, not an elevated one, for the safest no-admin Windows installation path.
- Install into `%LOCALAPPDATA%\Programs\Ollama` with the official PowerShell installer script and then reopen PowerShell so PATH updates are visible.
- Verify the CLI with `ollama --version`, start `ollama serve` if the background service is not already active, and confirm `http://127.0.0.1:11434/api/tags` responds.
- Pull the default repository model tags exactly as written if you want the shipped config and agent templates to work unchanged.

### No-admin Ollama commands and default model pulls

```
$env:OLLAMA_INSTALL_DIR = "$env:LOCALAPPDATA\Programs\ollama"
irm https://ollama.com/install.ps1 | iex
ollama --version
ollama serve
Invoke-WebRequest http://127.0.0.1:11434/api/tags -UseBasicParsing
ollama pull qwen3-embedding:8b
ollama pull kimi-k2.6:cloud
ollama pull qwen3.5:cloud
ollama pull gpt-oss:120b-cloud
ollama pull qwen3.5:397b-cloud
ollama pull llama3.2-vision:11b
```



## 5. Runtime, Release, and Operator Diagnostics

### Running the application

- Development server: ``python Tlamatini/manage.py runserver --noreload``.
- Preferred async/dev bootstrap: ``python Tlamatini/manage.py startserver``, which starts MCP services before the Django server.
- Production-style ASGI entrypoint: ``daphne -b 127.0.0.1 -p 8000 tlamatini.asgi:application``.
- Current startup also re-applies GPU-performance / Ollama-pinning hooks in the background on supported NVIDIA Windows hosts, so restart-time behavior stays closer to the tuned development baseline.
- That same early startup window is also where a staged ``DB/ToLoad/db.sqlite3`` file is promoted into the live database path, before Django opens SQLite.
- In frozen mode, startup now also self-heals the per-user Windows Installed-apps entry when ``Uninstaller.exe`` is present beside ``Tlamatini.exe``, so old installs can gain a standard uninstall surface without a reinstall.
- Startup now also prints a ``--- [VERSION] Tlamatini ...`` banner, making the running build visible in both the console and ``tlamatini.log`` without an HTTP call.
- Startup cleans pool state, repopulates the agent registry, launches MCP metrics/file-search servers, and then serves HTTP plus WebSocket traffic.

### Embedding-memory pre-flight guard

- README and BookOfTlamatini now document an embedding-memory pre-flight guard for GPU hosts before a directory-context load starts its FAISS embedding burst.
- On supported NVIDIA hosts it estimates embedding-model VRAM pressure, and when the projected load is too high it emits a non-blocking warning chat bubble instead of silently letting RAM<->VRAM thrash surprise the operator.
- CPU-only, AMD, and Apple-Silicon hosts stay fail-open: the guard becomes a no-op when the NVIDIA probe does not apply.
- The practical operator response is straightforward: switch to a smaller embedding model, reconnect if needed, or proceed knowingly with the heavier model.

### Reconnect and restart safeguards

- Config -> Models and Config -> URLs dialogs now track their pre-edit baseline and can show a reconnect-required dialog when the saved values change what the live chat session should trust.
- The restored-session autoload path now buffers early WebSocket frames so context-loading spinners and disabled-input state are not lost during automatic reconnect/restore flows.
- Startup and restart behavior now also re-apply GPU performance and Ollama keep-alive hooks in the background on supported NVIDIA Windows hosts, improving warm-model readiness without blocking Django boot.
- Browser-driven attention routing is now part of that operator-safety layer too: Ask Execs prompts and Notifier events can raise a taskbar flash plus a log banner without relying on the browser window already being visible.
- Windows process hygiene is now part of that safety story too: detached no-window spawns and the three-tier orphan reaper reduce the chance that Task Manager shows stale Tlamatini-icon console helpers after long runs.
- Ask Execs extends that safety story into execution approval itself: the operator can now stop a destructive chain before the next mutation instead of only auditing it after the fact in Exec Report.

### Prompt catalog and answer readability discipline

- Version ``1.3.2`` tightened the HTML answer contract with a Prime Directive on visual readability: explicit background and text color, no grey-on-dark body text, and safer table-body defaults.
- The seeded ``Prompts`` dropdown was also re-sorted into a learner path: context-only Q&A first, then metrics, files search, shell, code generation, vision, specialized single-tool actions, agent control, Unreal, and heavier Multi-Turn/ACPX demos last.

- Those readability rules remain in force in the current documentation set, and the newer `v1.12.0` release state keeps the version badge, runtime surfaces, self-knowledge wording, STM32er/ESP32er demo prompts, and operator handbook aligned.

## Ask Execs and execution approval

- Introduced in `v1.10.0` and still part of the current `v1.12.0` surface, `Ask Execs` is the Multi-Turn-only safety modifier that makes Tlamatini ask before each state-changing Tool, MCP, wrapped agent, or skill-backed execution instead of running it immediately.
- The permission dialog is explicit and auditable: it names the Tool or Agent family, the underlying raw tool name, the full parameters, the program or command to be executed, and the shell or execution surface involved.
- Proceed runs that one step and then prompts again at the next state-changing step; Deny halts the entire chain immediately and appends a red `Execution interrupted` banner even when Exec Report itself is off.
- Under the hood, `agent/exec\_permission.py` provides `ExecPermissionBroker`, which lets the synchronous worker-thread executor emit a permission request onto the WebSocket event loop and then block on a `threading.Event` until the browser replies.
- The gate sits after deduplication and quota checks, so skipped calls never prompt and denied calls never appear as executed rows; only work that truly ran lands in Exec Report.
- The round-trip is fail-safe: browser disconnect, emit failure, cancel, or broker shutdown all resolve to Deny, so an unconfirmed state-changing action never slips through just because the UI vanished at the wrong time.

## Windows attention issuing

- The current Windows attention path is explicit and local: the browser calls `POST /agent/flash\_window/`, and the backend routes that request through `agent/window\_flash.py`.
- That helper can flash the `Tlamatini.exe` console/taskbar window with `FlashWindowEx` and always prints an uppercase attention banner, so the signal survives in `tlamatini.log` even when the browser is minimized.
- Two concrete reasons are wired today: Ask Execs execution approval prompts and Notifier notifications. The path is best-effort and fail-safe, degrading cleanly on non-Windows or windowless launches.

## Unreal MCP runtime behavior

- Each Unreal run is one command: load `config.yaml`, open a fresh TCP socket, send `{"type": "command", "params": {"params": {}}`, read until valid JSON arrives, and log one atomic `INI\_SECTION\_UNREALER<<< ... >>>END\_SECTION\_UNREALER` block.
- The wrapped-tool path stores chat runs under `agent/agents/pools/\_chat\_runs/\_unrealer\_<seq>\_<id>/`, while visual flows use normal `unrealer\_<n>` pool folders and can chain response fields through Parametrizer mappings.
- Exec Report treats Unreal as its own row family, and troubleshooting stays concrete: connection-refused means the plugin is not listening, while read-timeout usually means UE5's game thread is busy.

## Orphan reaper runtime behavior

- Tlamatini now ships a three-tier orphan reaper in `Tlamatini/agent/orphan\_reaper.py` focused on Windows console-host leftovers such as `conhost.exe` and `openconsole.exe`.
- Tier 1 runs after spawn-capable Multi-Turn tool calls, Tier 2 runs once after the final answer in a background thread, and Tier 3 runs again during application shutdown through the same cleanup path that already tears down pools.
- The candidate set stays narrow on purpose: dead descendants of the current process tree, orphaned console hosts tied to that tree, and pool-linked processes whose `cmdline` still points into `agents/pools/...` even though tracking records are gone.
- Prevention landed alongside cleanup: Windows spawn sites now use `CREATE\_NO\_WINDOW | DETACHED\_PROCESS | CREATE\_NEW\_PROCESS\_GROUP` with stdio piped to `DEVNULL`, so the console host is usually never allocated in the first place.
- The ACPX runtime now kills full process trees instead of only top-level wrappers, and pool-agent scripts gained a conservative `subprocess.Popen` guard so forgotten descendants still inherit `CREATE\_NO\_WINDOW`.



- If something survives both cleanup tiers, Tlmatini sends a second chat bubble listing each surviving `name + PID`; the reaper never raises into the user path because a cleanup crash would be worse than the leftovers it tried to remove.

## Release pipeline

- Release production is a three-step pipeline: `build.py` -> `build\_uninstaller.py` -> `build\_installer.py`.
- The final distributable is the full versioned release folder `dist/Tlmatini\_Release\_v<version>/`, not a stray executable copied outside its payload.
- Use `build.py --self-modify` when you intentionally want a release that ships `TlmatiniSourceCode/` so she can inspect or modify herself at runtime.
- Current `build.py` treats `README.md` and `jd-cli/` as required post-build assets and fails hard if those payloads are missing.
- Bundled support scripts cover shortcut creation/removal, `.flw` association, the PowerShell launcher, Windows-specific installer ergonomics, and the per-user Installed-apps registration path.

## Windows uninstall registration

- Introduced in `v1.11.0` and still carried by the current `v1.12.0` release, the frozen install now behaves like a real Windows application: `install.py` writes a per-user HKCU Add/Remove Programs entry so Tlmatini appears in Settings -> Apps -> Installed apps and in the legacy Programs and Features list.
- The entry carries `DisplayName`, `DisplayVersion`, `InstallLocation`, `DisplayIcon`, `UninstallString`, `QuietUninstallString`, `NoModify`, `NoRepair`, and best-effort `EstimatedSize`, all pointing at the bundled `Uninstaller.exe` without requiring administrator rights.
- The matching runtime self-heal in `agent/apps.py` calls `windows\_app\_registration.self\_heal\_for\_frozen()` on every frozen launch, so installs created before this feature existed can appear in Windows' uninstall UI after the next normal app start.

## Exec Report

- Exec Report is a Multi-Turn-only transparency layer that appends one operation table per state-changing agent family to the final answer.
- Rows are recorded from the live tool-call stream rather than guessed from the LLM prose, so the report is the operational ground truth.
- Each row receives a SUCCESS/FAILURE verdict from raw tool returns, making long installs, deployments, and remediations inspectable after the fact.
- When Ask Execs is enabled, Exec Report and the red denial banner complement each other: already-executed steps still render as tables, while the denied step stays out of the tables and is surfaced only through the interruption banner.

## ACPX and skills

- ACPX lets Tlmatini spawn external coding-agent CLIs such as Codex, Claude Code, Cursor, Gemini, Qwen, and others as managed child processes.
- It pairs those agents with markdown-driven `SKILL.md` packages, validated I/O contracts, permission gating, and append-only audit logs.
- Transport-aware drain rules and bounded event bodies reduce latency while protecting the LLM context budget during external-agent relays.
- The operator-facing references are `README.md` and `ACPX.md`, while the implementation lives under `agent/acpx/`, `agent/skills/`, and `agent/skills\_pkg/`.

## Application log

- The built-in `tlmatini.log` file captures both stdout and stderr through a tee stream initialized in `manage.py` before Django starts.
- In source mode the log sits next to `manage.py`; in frozen mode it lives next to the executable.

- Immediate flush behavior makes the log the primary forensic artifact for startup problems, warnings, tracebacks, and runtime diagnostics.

## 6. Agent Catalog and Runtime Model

Tlmatini currently exposes 69 workflow-agent templates.

| Category               | Representative agents                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Control                | starter, ender, stopper, cleaner, barrier, flowbacker                                                                      |
| Execution and files    | executer, pythonxer, pser, file_creator, file_extractor, file_interpreter, de_compressor, playwrighter, windower, unrealer |
| DevOps and infra       | gitter, dockerer, kubernetes, jenkinser, ssher, scper                                                                      |
| Data and APIs          | sqler, mongoxer, apirer, crawler, googler                                                                                  |
| Monitoring and routing | monitor_log, monitor_netstat, flowhypervisor, forker, asker, counter, and, or                                              |
| Communication          | notifier, emailer, recmailer, telegramer, telegramrx, teletlmatini, whatsapper, whatstlmatini                              |
| Security and media     | kyber_keygen, kyber_cipher, kyber_decipher, image_interpreter, shoter, j_decompiler                                        |
| Workflow intelligence  | flowcreator, gatewayer, gateway_relayer, node_manager, parametrizer, prompter, summarizer, acpxer                          |

All workflow agents follow a common deployment pattern: template directory, YAML configuration, session-scoped pool copy, PID/status/log files, target/source wiring, and optional reanimation state.

### Agent catalog validation

- The authoritative human-readable source for workflow-agent descriptions is `agents\_descriptions.md` at the repo root, not an embedded JavaScript map or a hard-coded Django list.
- Current validation compares live template directories, `agents\_descriptions.md` rows, and the README / Book bestiary counts so the generated dossier stays aligned with the running product.
- The ACP sidebar hover tooltip and the right-click Description dialog both resolve through that markdown file first; `README.md` is only a legacy fallback when the dedicated description file is absent.

### How agent runtimes are shaped

- Every workflow agent follows the same operational skeleton: template directory, `config.yaml`, a session-scoped pool copy, PID/status/log files, and explicit source/target wiring.
- Chat-wrapped tool calls launch isolated runtime copies under `agent/agents/pools/\_chat\_runs\_/\_`, while ACP uses named pool folders such as `starter\_1` or `unrealer\_1`.
- Specialized agents now stretch the platform in different directions: ACPXer drives external coding-agent CLIs, Kalier drives a remote or tunneled Kali Linux tool server, STM32er drives a zero-config STM32 firmware MCP bridge, Unrealer drives a live UE5 editor, and TeleTlmatini / WhatsTlmatini bridge full Tlmatini conversations into messaging platforms.

### Reviewer and Analyzer spotlight

- The workflow catalog now includes two new high-value specialists: Reviewer and Analyzer, lifting the canvas inventory to 64 agents and the seed-skill catalog to 23 SKILL.md packages.
- Reviewer is LLM-powered: it resolves a git diff for `repo\_path`, reviews it with a senior-engineer prompt, and emits an `INI\_SECTION\_REVIEWER` block whose first routable field is `verdict = APPROVE | REQUEST\_CHANGES | COMMENT`.
- Analyzer is deterministic and scanner-driven: it runs whichever of `bandit`, `semgrep`, `ruff`, `eslint`, `gitleaks`, and `pip-audit` are installed on PATH over `target\_path`, then emits an `INI\_SECTION\_ANALYZER` block with `status` and `total\_findings` for downstream routing.
- Both agents always trigger `target\_agents`, so a downstream Forker can branch on `{verdict}`, `{status}`, or `{total\_findings}` instead of scraping prose.
- They are intentionally canvas-only workflow agents: there is no wrapped `chat\_agent\_reviewer` or `chat\_agent\_analyzer`, and therefore no duplicate Exec Report row family for those names.
- The chat-side counterparts live in the ACPX skill catalog instead: `code-review` exposes senior-engineer git-diff review, while `security-audit` exposes the deterministic multi-scanner sweep.

## Operator surface counts

- The live operator surface now stands at 69 workflow agents, 76 Multi-Turn tools, 12 ACPX tools, and 26 skills.
- Source inspection confirms the newer total: 44 wrapped chat-agent tools in ``chat_agent_registry.py``, which combines with 20 core Python tools and 12 ACPX/Skill tools for 76 Multi-Turn tools overall.
- The workflow-agent and wrapped-tool totals align cleanly with the handbooks, while the skill total is validated from the on-disk ``agent/skills_pkg/`` catalog so the dossier stays honest even when a simplified markdown summary lags behind the live tree.
- This matters operationally because the planner never binds everything at once: the documented default ``max_selected_tools`` cap stays at 20, so breadth of capability does not mean uncontrolled tool sprawl per turn.

## Kalier spotlight

- Kalier remains the 67th workflow agent in ``v1.7.1``, but its newest upgrade is usability: Tlmatini now acts as the embedded MCP-Kali-Server client for chat-side runs.
- It can issue one capability per run, including ``nmap``, ``gobuster``, ``dirb``, ``nikto``, ``sqlmap``, ``metasploit``, ``hydra``, ``john``, ``wpscan``, ``enum4linux``, arbitrary ``command``, or a safe ``health`` probe of the remote server.
- The agent emits ``INI_SECTION_KALIER`` with ``action``, ``endpoint``, ``subject``, ``return_code``, ``success``, ``timed_out``, and ``server_url``, so downstream Forker or Parametrizer logic can branch on results without scraping prose.
- The practical operator result is simpler prompts: after the Kali box URL is configured once, normal Multi-Turn requests no longer repeat ``server_url`` unless you intentionally override it for a one-off target.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_kalier`` now auto-injects the configured ``kali_server_url``, while the visual Kalier canvas node still stores ``server_url`` explicitly in YAML per node.
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- Before any compile-or-hardware action, STM32er runs a critical-mission fail-safe preflight over the arm-none-eabi toolchain, STM32CubeIDE, programmer path, ST-LINK probe, and target family; compile-only steps can run boardless, but flash, erase, reset, serial, and SWD/live-memory operations are refused when the environment or device is wrong.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_stm32er`` takes one ``action`` per call, while the visual STM32er canvas node stores the same fields in YAML and triggers downstream agents on both success and failure.
- The tool surface includes the full project lifecycle plus hardware-in-the-loop composites: ``validate``, ``bootstrap``, ``create_project``, ``write_source``, ``build``, ``build_and_flash``, ``serial_session``, ``live_monitor``, and the rest of the 23 MCP verbs, with every run emitting an ``INI_SECTION_STM32ER`` block for Forker or Parametrizer routing.
- Config -> URLs now seeds the chat path with ``stm32_mcp_server_script``, ``stm32_mcp_python``, ``stm32_template_dir``, ``stm32_ide_root``, ``stm32_mcp_repo_url``, and ``stm32_mcp_install_dir``, so firmware prompts normally describe only the task and target board instead of the plumbing.

## Windower spotlight

- Windower is the new 66th workflow agent: a deterministic Win32 window manager that finds an application window by title and performs one lifecycle operation on the window itself instead of clicking inside it.

- It supports ``focus``, ``minimize``, ``maximize``, ``restore``, ``move``, ``resize``, ``move_resize``, ``close``, ``topmost``, ``untopmost``, ``arrange``, and ``list``, with ``substring``, ``exact``, or ``regex`` title matching plus ``match_index`` for duplicate titles.
- The agent emits ``INI_SECTION_WINDOWER`` with ``action``, ``window_title``, ``matched``, ``match_count``, ``state``, ``left``, ``top``, ``width``, and ``height``, so downstream Forker or Parametrizer logic can branch on presence, state, or geometry.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_windower`` accepts free-form key=value requests, while the visual Windower canvas node stores the same operation fields in YAML.
- Windower is the desktop-window sibling of Mouser and Keyboarder: use Windower when the goal is the window as a whole, Mouser for controls inside it, and Keyboarder for text entry into it.
- Because Windower changes real window state, its executions appear in Exec Report and it always triggers ``target_agents`` whether the action succeeds or fails.

## Playwrighter spotlight

- Playwrighter is the new 65th workflow agent in ``v1.5.0``: a deterministic Playwright-powered browser automator for Chromium, Firefox, or WebKit that executes ordered interactive steps instead of static fetches.
- It covers the gap between Crawler and Googler: logins, multi-step forms, wizard clicks, JS-rendered SPA scraping, screenshots after interaction, assertions, downloads, and authenticated end-to-end UI checks.
- The agent emits ``INI_SECTION_PLAYWRIGHTER`` with ``start_url``, ``final_url``, ``status``, ``steps_run``, ``assert_result``, and extracted values so downstream Forker or Parametrizer logic can branch on pass/fail or reuse scraped data.
- Two operator surfaces ship in lock-step: the wrapped Multi-Turn tool ``chat_agent_playwrighter`` takes the whole script as ``steps_json``, while the visual Playwrighter canvas node stores the same declarative step list in YAML.
- Session continuity is built in: ``headless: false`` lets operators watch it drive, and ``storage_state_in`` / ``storage_state_out`` carry login state across runs without forcing a manual browser setup each time.
- Because Playwrighter is state-changing, its executions appear in Exec Report and it always triggers ``target_agents`` whether the run succeeds or fails.

## Reviewer precision spotlight

- The ``v1.4.1`` Reviewer refinement tightened accuracy rather than adding a new surface: when ``diff_ref`` is empty, the review prompt labels the diff as the uncommitted working tree plus staged area, so the model must not describe those findings as already committed or pushed.
- The same patch teaches the model Tlmatini's managed-secret convention: ``agent/config.json`` and selected ``agent/agents/*/config.yaml`` files can hold live local credentials in a keyed working copy, while ``regen_secrets.py --mode push-able`` scrubs them back to placeholders before commit.
- That guidance is mirrored into the ``code-review`` SKILL.md package too, keeping the chat-surface review behavior and the canvas Reviewer agent aligned on commit-state wording and secret-severity expectations.

## Native-dialog spotlight

- The current tagged release ``v1.4.2`` removes Tkinter from the unstable runtime-facing dialog path and replaces it with ``Tlmatini/agent/native_dialogs.py``, a native Windows dialog bridge used by browser-triggered pickers.
- This change pairs with the existing DB and operator dialogs: file and folder selection still feels local and GUI-first, but the fragile Tkinter dependency is no longer part of the interactive runtime path that users trigger from chat or ACP surfaces.
- The patch arrived with dedicated tests (``test_native_dialogs.py``) and with follow-up orphan-reaper/runtime adjustments, so the release reads as a stability pass rather than a cosmetic refactor.

## Unrealer spotlight

- Unreal MCP is a UE5 plugin that runs inside the editor and listens on ``127.0.0.1:55557`` for one JSON command per TCP connection; Tlmatini is the client side of that link, not the plugin host.
- The Unrealer workflow agent forwards any command the connected plugin build exposes, up to a 53-command surface across nine categories: actor manipulation (incl. viewport screenshots), Blueprint authoring and node wiring, input mappings, UMG widget building, in-editor Python/console execution, level I/O, asset import, and material authoring. Headless build/cook/test is out of scope (it needs UnrealEditor-Cmd, not the editor socket).

- The same integration is available in both operator surfaces: checked Multi-Turn via ``chat_agent_unrealer``, and ACP canvas flows via the visual Unrealer node plus Parametrizer chaining.
- Each Unrealer run is one command: load ``config.yaml``, open a fresh TCP socket, send ``{"type": "command", "params": params}``, read until valid JSON arrives, and log one atomic ``INI_SECTION_UNREALER<<< ... >>>END_SECTION_UNREALER`` block.
- The wrapped-tool path stores chat runs under ``agent/agents/pools/_chat_runs/_unrealer_<seq>_<id>/_``, while visual flows use normal ``unrealer_<n>`` pool folders and can chain response fields through Parametrizer mappings.
- Exec Report treats Unrealer as its own row family, and troubleshooting stays concrete: connection-refused means the plugin is not listening, while read-timeout usually means UE5's game thread is busy.

## De-Compressor spotlight

- De-Compressor is the deterministic archive worker for compression and decompression tasks, deciding direction from whichever side exposes a recognized archive extension.
- Supported archive families are ``.gz``, ``.zip``, ``.7z``, ``.tar.gz``, and ``.gz.tar``; file-to-folder extraction and file-or-directory packing are both documented in the README and Book.
- Password handling is explicit: ``passwordless=true`` skips it, while ``passwordless=false`` requires the ``DE_COMPRESSER_PWD`` environment variable and fails fast if the secret is missing.
- The agent is reachable from both operator surfaces: ACP canvas nodes wire through dedicated connection-update views, and checked Multi-Turn can invoke it through ``chat_agent_de_compressor``.
- Format engines are practical rather than magical: `stdlib`gzip`` and ``zipfile`` cover core cases, ``.7z`` is preferred for encrypted ``.7z`` or ``.zip``, and ``py7zr`` was added to ``requirements.txt`` as the Python fallback.
- Every run emits an ``INI_SECTION_DE_COMPRESSER<<< ... >>>END_SECTION_DE_COMPRESSER`` block and still triggers ``target_agents``, so downstream Parametrizer or Raiser logic can branch on ``success=true|false`` instead of guessing from prose.

## Orphan-reaper spotlight

- Tlamatini now ships a three-tier orphan reaper in ``Tlamatini/agent/orphan_reaper.py`` focused on Windows console-host leftovers such as ``conhost.exe`` and ``openconsole.exe``.
- Tier 1 runs after spawn-capable Multi-Turn tool calls, Tier 2 runs once after the final answer in a background thread, and Tier 3 runs again during application shutdown through the same cleanup path that already tears down pools.
- The candidate set stays narrow on purpose: dead descendants of the current process tree, orphaned console hosts tied to that tree, and pool-linked processes whose ``cmdline`` still points into ``agents/pools/...`` even though tracking records are gone.

## 7. Repository Facts and Git Changes

| Metric                                             | Value  |
|----------------------------------------------------|--------|
| Tracked files in git                               | 519    |
| Workflow agents                                    | 69     |
| Multi-Turn tools                                   | 76     |
| Wrapped chat-agent tools                           | 44     |
| Skills                                             | 26     |
| agents_descriptions.md rows                        | 69     |
| Django migrations                                  | 107    |
| Frontend JavaScript modules                        | 28     |
| Frontend CSS files                                 | 6      |
| HTML templates                                     | 4      |
| Python requirements                                | 71     |
| Binary/asset tracked files skipped from line count | 22     |
| Resolved version                                   | 1.12.0 |
| Version source                                     | git    |

### Latest commits

| Date       | Commit  | Subject                                                                                       |
|------------|---------|-----------------------------------------------------------------------------------------------|
| 2026-05-31 | 1df130b | Adding ES32er Agent programmer, debugger and uploader, by angelahack1, to push version 1.12.0 |
| 2026-05-31 | 373f6d8 | Implementing visual windows attention issuing, by angelahack1                                 |
| 2026-05-30 | 5ec9593 | Delete CNAME                                                                                  |
| 2026-05-30 | e716c5d | Updating visual documentation to version 1.11.0, by angelahack1                               |
| 2026-05-30 | 0d39ec0 | Setting documentatioon as v1.11.0, by angelahack1                                             |
| 2026-05-30 | 1b37aad | Fixing commit to asking method.                                                               |
| 2026-05-30 | fa1eb26 | Refined and tested at runtime the ask mechanism execution, by Tlamatini's-AutoBot.            |
| 2026-05-30 | 36fd045 | Create CNAME                                                                                  |
| 2026-05-29 | a300710 | Refined at runtime the ask mechanism execution, by Tlamatini's-AutoBot.                       |
| 2026-05-29 | b0efaa9 | Refined the ask mechanism execution, by Tlamatini's-AutoBot.                                  |

### Changes since the last committed PDF/PPTX refresh

Last committed visual-dossier refresh: 606421d on 2026-05-31 — Getting rid of the native toast in documentation, by angelahack1

- Since the last committed PDF/PPTX refresh, `v1.12.0` added ESP32er as the 69th workflow agent and `chat\_agent\_esp32er` as the new embedded-firmware surface: she now drives PlatformIO Core directly to scaffold, build, upload, and monitor ESP32-class firmware with zero-config bootstrap and a serial-aware safety preflight.

### Visual-dossier change appendix 1 of 1

| Date       | Commit  | Subject                                                                                       |
|------------|---------|-----------------------------------------------------------------------------------------------|
| 2026-05-31 | 1df130b | Adding ES32er Agent programmer, debugger and uploader, by angelahack1, to push version 1.12.0 |

### Git changes from today



- Today's headline change is `v1.12.0` ESP32er: Tlamatini now reaches 69 workflow agents and adds a direct PlatformIO Core bridge, so she can scaffold, author, build, upload, and monitor ESP32-class firmware from chat or canvas without an external MCP server.
- Today's headline change is the new versioning system: SemVer policy, git-tag sourcing, runtime version surfaces, and build-time embedding across the Windows artefacts.
- Operator attention routing also changed: when Ask Execs or a Notifier event needs the user, Tlamatini can now flash her own Windows taskbar presence and write an uppercase attention banner into `tlamatini.log`.
- Documentation itself changed today, so the regenerated dossier is part of the tracked operator surface rather than an external afterthought.

## Today's commit appendix 1 of 1

| Date       | Commit  | Subject                                                                                       |
|------------|---------|-----------------------------------------------------------------------------------------------|
| 2026-05-31 | 1df130b | Adding ES32er Agent programmer, debugger and uploader, by angelahack1, to push version 1.12.0 |
| 2026-05-31 | 606421d | Getting rid of the native toast in documentation, by angelahack1                              |
| 2026-05-31 | 373f6d8 | Implementing visual windows attention issuing, by angelahack1                                 |

## 8. Effective Line Inventory by Language

Methodology: tracked text files only. Blank lines and comment-only lines are excluded. Python counts also remove module, class, and function docstrings detected through AST parsing.

| Language          | Files | Effective lines | Total lines | Share |
|-------------------|-------|-----------------|-------------|-------|
| Python            | 302   | 80,392          | 112,417     | 68.8% |
| Markdown          | 61    | 13,859          | 17,694      | 11.9% |
| JavaScript        | 28    | 13,515          | 17,040      | 11.6% |
| CSS               | 6     | 3,600           | 4,565       | 3.1%  |
| YAML              | 73    | 1,552           | 3,415       | 1.3%  |
| HTML              | 4     | 858             | 876         | 0.7%  |
| Other text        | 3     | 791             | 970         | 0.7%  |
| Prompt template   | 2     | 685             | 801         | 0.6%  |
| PowerShell        | 5     | 508             | 754         | 0.4%  |
| JavaScript module | 1     | 401             | 470         | 0.3%  |
| JSON              | 5     | 395             | 395         | 0.3%  |
| Text              | 5     | 188             | 218         | 0.2%  |
| Protocol Buffers  | 1     | 20              | 33          | 0.0%  |
| Batch             | 1     | 13              | 24          | 0.0%  |

**Total effective lines: 116,777**

## 9. Largest Effective Source Files

| Path                                                      | Language   | Effective | Total  |
|-----------------------------------------------------------|------------|-----------|--------|
| Tlmatini/agent/views.py                                   | Python     | 7,360     | 10,677 |
| Tlmatini/agent/tests.py                                   | Python     | 4,990     | 6,406  |
| Tlmatini/agent/tools.py                                   | Python     | 2,581     | 3,718  |
| Tlmatini/agent/doc_generation/complete_project_docs.py    | Python     | 2,180     | 2,445  |
| BookOfTlmatini.md                                         | Markdown   | 2,095     | 2,948  |
| Tlmatini/agent/agents/flowcreator/agentic_skill.md        | Markdown   | 1,985     | 2,200  |
| Tlmatini/agent/static/agent/css/agentic_control_panel.css | CSS        | 1,731     | 2,313  |
| Tlmatini/agent/static/agent/js/agent_page_init.js         | JavaScript | 1,435     | 1,687  |
| Tlmatini/agent/static/agent/css/agent_page.css            | CSS        | 1,308     | 1,592  |
| Tlmatini/agent/static/agent/js/acp-canvas-core.js         | JavaScript | 1,307     | 1,617  |
| Tlmatini/agent/consumers.py                               | Python     | 1,300     | 1,638  |
| README.md                                                 | Markdown   | 1,286     | 1,829  |
| Tlmatini/agent/static/agent/js/acp-agent-connectors.js    | JavaScript | 1,235     | 1,382  |
| Tlmatini/agent/test_stm32er_agent.py                      | Python     | 1,200     | 1,707  |
| Tlmatini/agent/agents/stm32er/stm32er.py                  | Python     | 1,191     | 1,642  |
| Tlmatini/agent/static/agent/js/agent_page_chat.js         | JavaScript | 1,184     | 1,622  |
| Tlmatini/agent/agents/whatstlmatini/whatstlmatini.py      | Python     | 1,059     | 1,323  |
| Tlmatini/agent/static/agent/js/acp-control-buttons.js     | JavaScript | 1,036     | 1,344  |
| Tlmatini/agent/acpx/tests.py                              | Python     | 1,035     | 1,299  |
| KIMI.md                                                   | Markdown   | 1,013     | 1,228  |
| Tlmatini/agent/chat_agent_registry.py                     | Python     | 978       | 1,000  |
| Tlmatini/agent/agents/teletlmatini/teletlmatini.py        | Python     | 903       | 1,170  |
| Tlmatini/agent/mcp_agent.py                               | Python     | 901       | 1,412  |
| Tlmatini/agent/static/agent/js/canvas_item_dialog.js      | JavaScript | 898       | 1,165  |
| Tlmatini/agent/agents/crawler/crawler.py                  | Python     | 896       | 1,285  |

# 10. Complete Tracked File Tree (Repository Appendix)

## Tree appendix 1 of 9

```
Tlamatini/
|-- .gitignore
|-- _acpx_pipeline_demo_report.md
|-- ACPX.md
|-- agents_descriptions.md
|-- BookOfTlamatini.md
|-- build.py
|-- build_installer.py
|-- build_uninstaller.py
|-- Claude-Desktop-KALI-MCP-Session.md
|-- CLAUDE.md
|-- CreateShortcut.json
|-- CreateShortcut.ps1
|-- eslint.config.mjs
|-- example_of_prompt_to_make_de-compressor_agent.txt
|-- install.py
|-- KIMI.md
|-- LICENSE
|-- OllamaPricing.png
|-- package.json
|-- PIVOT_CHANGES.md
|-- pnpm-lock.yaml
|-- prompt_example_2_create_agent.txt
|-- README.md
|-- regen_secrets.py
|-- register_flw.ps1
|-- RemoveShortcut.ps1
|-- requirements.txt
|-- Tlamatini-Kali-Setup.md
|-- Tlamatini.ico
|-- Tlamatini.jpg
|-- Tlamatini.ps1
|-- tlamatini_app_summary.pdf
|-- Tlamatini_eXtended_Artificial_Intelligence_Humanly_Tempered.pptx
|-- uninstall.py
|-- unregister_flw.ps1
|-- VERSIONING.md
|-- versioning.py
|-- .codex/
|   |-- skills/
|   |   |-- full-project-pdf-dossier/
|   |   |   |-- SKILL.md
|   |   |-- overlap-safe-pptx-dossier/
|   |   |   |-- SKILL.md
|   |-- .github/
|   |   |-- workflows/
|   |   |   |-- name-guard.yml
|   |-- docs/
|   |   |-- claude/
|   |   |   |-- acpx.md
|   |   |   |-- agents.md
|   |   |   |-- architecture.md
|   |   |   |-- exec-report.md
|   |   |   |-- frontend.md
|   |   |   |-- gotchas.md
|   |   |   |-- INDEX.md
|   |   |   |-- mcp-tools.md
|   |   |   |-- multi-turn.md
|   |   |   |-- recent-fixes.md
|   |-- pyinstaller_hooks/
|   |   |-- hook-numpy.py
|-- Tlamatini/
|   |-- cat_art.py
|   |-- manage.py
|   |-- .agents/
|   |   |-- workflows/
|   |   |   |-- create_new_agent.md
|   |-- .mcps/
|   |   |-- create_new_mcp.md
|   |-- agent/
|   |   |-- __init__.py
|   |   |-- admin.py
|   |   |-- apps.py
|   |   |-- capability_registry.py
|   |   |-- chain_files_search_lcel.py
|   |   |-- chain_system_lcel.py
```

## Tree appendix 2 of 9

```
-- chat_agent_registry.py
-- chat_agent_runtime.py
-- chat_history_loader.py
-- command_togen_pb2.txt
-- config.json
-- config_loader.py
-- constants.py
-- consumers.py
-- embedding_memory_guard.py
-- exec_permission.py
-- filesearch.proto
-- filesearch_pb2.py
-- filesearch_pb2_grpc.py
-- global_execution_planner.py
-- global_state.py
-- gpu_perf.py
-- inet_determiner.py
-- mcp_agent.py
-- mcp_files_search_client.py
-- mcp_files_search_server.py
-- mcp_system_client.py
-- mcp_system_server.py
-- models.py
-- native_dialogs.py
-- orphan_reaper.py
-- path_guard.py
-- prompt.pmt
-- rag_enhancements.py
-- routing.py
-- test_build_scripts.py
-- test_de_compressor.py
-- test_embedding_memory_guard.py
-- test_esp32er_agent.py
-- test_flow_contracts.py
-- test_kalier_agent.py
-- test_native_dialogs.py
-- test_orphan_reaper.py
-- test_password_quoting.py
-- test_playwrighter_agent.py
-- test_stm32er_agent.py
-- test_unrealer_agent.py
-- test_windower_agent.py
-- test_windows_app_registration.py
-- tests.py
-- Tlamatini.md
-- tools.py
-- urls.py
-- version.py
-- views.py
-- web_search_llm.py
-- window_flash.py
-- windows_app_registration.py
-- acpx/
|  -- __init__.py
|  -- agent_registry.py
|  -- config.py
|  -- permissions.py
|  -- runtime.py
|  -- service.py
|  -- session_store.py
|  -- tests.py
|  -- tools.py
|  -- windows_spawn.py
-- agents/
|  -- acpxer/
|  |  -- acpxer.py
|  |  -- config.yaml
|  -- analyzer/
|  |  -- analyzer.py
|  |  -- config.yaml
|  -- and/
|  |  -- and.py
|  |  -- config.yaml
|  -- apirer/
|  |  -- apirer.py
|  |  -- config.yaml
```

## Tree appendix 3 of 9

```

|-- asker/
|   |-- asker.py
|   |-- config.yaml
|-- barrier/
|   |-- barrier.py
|   |-- config.yaml
|   |-- test_barrier.py
|-- cleaner/
|   |-- __init__.py
|   |-- cleaner.py
|   |-- config.yaml
|-- counter/
|   |-- config.yaml
|   |-- counter.py
|-- crawler/
|   |-- config.yaml
|   |-- crawler.py
|-- croner/
|   |-- config.yaml
|   |-- croner.py
|-- de_compressor/
|   |-- config.yaml
|   |-- de_compressor.py
|-- deleter/
|   |-- config.yaml
|   |-- deleter.py
|-- dockerer/
|   |-- config.yaml
|   |-- dockerer.py
|-- emailer/
|   |-- config.yaml
|   |-- emailer.py
|-- ender/
|   |-- config.yaml
|   |-- ender.py
|-- esp32er/
|   |-- config.yaml
|   |-- esp32er.py
|-- executor/
|   |-- config.yaml
|   |-- executor.py
|-- file_creator/
|   |-- config.yaml
|   |-- file_creator.py
|-- file_extractor/
|   |-- config.yaml
|   |-- file_extractor.py
|-- file_interpreter/
|   |-- config.yaml
|   |-- file_interpreter.py
|-- flowbacker/
|   |-- config.yaml
|   |-- flowbacker.py
|-- flowcreator/
|   |-- agentic_skill.md
|   |-- config.yaml
|   |-- flowcreator.py
|-- flowhypervisor/
|   |-- config.yaml
|   |-- flowhypervisor.py
|   |-- hypervisor_alert.wav
|   |-- monitoring-prompt.pmt
|-- forker/
|   |-- config.yaml
|   |-- forker.py
|-- gateway_relayer/
|   |-- config.yaml
|   |-- gateway_relayer.md
|   |-- gateway_relayer.py
|-- gatewayer/
|   |-- config.yaml
|   |-- gatewayer.md
|   |-- gatewayer.py
|   |-- gatewayer_explanation.md
|-- gitter/
|   |-- config.yaml

```

## Tree appendix 4 of 9

```
| | | `-- gitter.py
| | | -- googler/
| | | | -- config.yaml
| | | | `-- googler.py
| | | -- image_interpreter/
| | | | -- config.yaml
| | | | `-- image_interpreter.py
| | | -- j_decompiler/
| | | | -- config.yaml
| | | | `-- j_decompiler.py
| | | -- jenkinser/
| | | | -- config.yaml
| | | | `-- jenkinser.py
| | | -- kalier/
| | | | -- config.yaml
| | | | `-- kalier.py
| | | -- keyboarder/
| | | | -- config.yaml
| | | | `-- keyboarder.py
| | | -- kubernetes/
| | | | -- config.yaml
| | | | `-- kubernetes.py
| | | -- kyber_cipher/
| | | | -- config.yaml
| | | | `-- kyber_cipher.py
| | | -- kyber_decipher/
| | | | -- config.yaml
| | | | `-- kyber_decipher.py
| | | -- kyber_keygen/
| | | | -- config.yaml
| | | | `-- kyber_keygen.py
| | | -- mongoxer/
| | | | -- config.yaml
| | | | `-- mongoxer.py
| | | -- monitor_log/
| | | | -- config.yaml
| | | | `-- monitor_log.py
| | | -- monitor_netstat/
| | | | -- config.yaml
| | | | `-- monitor_netstat.py
| | | -- mouser/
| | | | -- config.yaml
| | | | `-- mouser.py
| | | -- mover/
| | | | -- config.yaml
| | | | `-- mover.py
| | | -- node_manager/
| | | | -- config.yaml
| | | | -- node_manager.md
| | | | -- node_manager.py
| | | | -- nodes_example.json
| | | | `-- nodes_example.yaml
| | | -- notifier/
| | | | -- config.yaml
| | | | `-- notifier.py
| | | -- or/
| | | | -- config.yaml
| | | | `-- or.py
| | | -- parametrizer/
| | | | -- config.yaml
| | | | `-- parametrizer.py
| | | -- playwrighter/
| | | | -- config.yaml
| | | | `-- playwrighter.py
| | | -- prompter/
| | | | -- config.yaml
| | | | `-- prompter.py
| | | -- pser/
| | | | -- config.yaml
| | | | `-- pser.py
| | | -- pythonxer/
| | | | -- config.yaml
| | | | `-- pythonxer.py
| | | -- raiser/
| | | | -- config.yaml
| | | | `-- raiser.py
```



## Tree appendix 5 of 9

```
-- recmailer/
|   |-- config.yaml
|   |-- recmailer.py
-- reviewer/
|   |-- config.yaml
|   |-- reviewer.py
-- scper/
|   |-- config.yaml
|   |-- scper.py
-- shooter/
|   |-- config.yaml
|   |-- shooter.py
-- sleeper/
|   |-- config.yaml
|   |-- sleeper.py
-- sqler/
|   |-- config.yaml
|   |-- sqler.py
-- ssher/
|   |-- config.yaml
|   |-- ssher.py
-- starter/
|   |-- config.yaml
|   |-- starter.py
-- stm32er/
|   |-- config.yaml
|   |-- stm32er.py
-- stopper/
|   |-- config.yaml
|   |-- stopper.py
-- summarizer/
|   |-- config.yaml
|   |-- summarizer.py
-- telegramer/
|   |-- config.yaml
|   |-- telegramer.py
-- telegramrx/
|   |-- config.yaml
|   |-- telegramrx.py
-- teletlamatini/
|   |-- config.yaml
|   |-- teletlamatini.py
-- unrealer/
|   |-- config.yaml
|   |-- unrealer.py
-- whatsapper/
|   |-- config.yaml
|   |-- whatsapper.py
-- whatstlamatini/
|   |-- config.yaml
|   |-- whatstlamatini.py
-- windower/
|   |-- config.yaml
|   |-- windower.py
-- doc_generation/
|   |-- complete_project_docs.py
|   |-- markdown_to_pdf.py
|   |-- refresh_project_docs.py
|   |-- test_output.pdf
-- images/
|   |-- HiIamTlamatini.png
|   |-- mockup.jpg
|   |-- RealisticTlamatini.png
|   |-- ShouldersUoTlamatini.png
|   |-- Tlamatini AI Agentic Advisor Video.mp4
|   |-- TlamatiniAbout.jpg
|   |-- TlamatiniDrawings.png
|   |-- TlammatiniLetsMakeSomeMagic.mp4
|   |-- TorsoTlamatini.png
|   |-- XAIHT-Tlamatini.mp4
-- imaging/
|   |-- converter.py
|   |-- image_interpreter.py
|   |-- screenshot.py
-- management/
|   |-- __init__.py
```

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|-- commands/
|   |-- acpxLint.py
|   |-- startserver.py
|-- migrations/
|   |-- 0001_initial.py
|   |-- 0002_populate_db.py
|   |-- 0003_add_cleaner.py
|   |-- 0004_add_deleter.py
|   |-- 0005_add_executer.py
|   |-- 0006_add_notifier.py
|   |-- 0007_add_stopper.py
|   |-- 0008_add_recmailer.py
|   |-- 0009_add_whatsapp.py
|   |-- 0010_add_pythonxer.py
|   |-- 0011_add_asker.py
|   |-- 0012_repopulate_all_agents.py
|   |-- 0013_add_forker.py
|   |-- 0014_repopulate_all_agents.py
|   |-- 0015_add_shoter.py
|   |-- 0016_repopulate_all_agents.py
|   |-- 0017_add_ssh.py
|   |-- 0018_repopulate_all_agents.py
|   |-- 0019_add_scper.py
|   |-- 0020_repopulate_all_agents.py
|   |-- 0021_add_telegramrx.py
|   |-- 0022_repopulate_all_agents.py
|   |-- 0023_add_mongoxer.py
|   |-- 0024_repopulate_all_agents.py
|   |-- 0025_add_telegramer.py
|   |-- 0026_repopulate_all_agents.py
|   |-- 0027_add_sqler.py
|   |-- 0028_repopulate_all_agents.py
|   |-- 0029_add_prompter.py
|   |-- 0030_repopulate_all_agents.py
|   |-- 0031_add_flowcreator.py
|   |-- 0032_repopulate_all_agents.py
|   |-- 0033_add_gitter.py
|   |-- 0034_add_dockerer.py
|   |-- 0035_add_pser.py
|   |-- 0036_add_kuberneter.py
|   |-- 0037_add_apirer.py
|   |-- 0038_add_jenkinser.py
|   |-- 0039_add_crawler.py
|   |-- 0040_add_summarizer.py
|   |-- 0041_add_flowhypervisor.py
|   |-- 0042_add_mouser.py
|   |-- 0043_agentmessage_conversation_user.py
|   |-- 0044_add_counter.py
|   |-- 0045_add_file_interpreter.py
|   |-- 0046_add_image_interpreter.py
|   |-- 0047_add_gateway.py
|   |-- 0048_add_gateway_relayer.py
|   |-- 0049_add_node_manager.py
|   |-- 0050_add_file_creator.py
|   |-- 0051_add_file_extractor.py
|   |-- 0052_add_kyber_keygen.py
|   |-- 0053_add_kyber_cipher.py
|   |-- 0054_add_kyber_decipher.py
|   |-- 0055_fix_kyber_cipher_names.py
|   |-- 0056_add_parametrizer.py
|   |-- 0057_add_flowbacker.py
|   |-- 0058_add_barrier.py
|   |-- 0059_add_j_decompiler.py
|   |-- 0060_add_agent_parametrizer_tool.py
|   |-- 0061_add_agent_starter_tool.py
|   |-- 0062_sync_agent_control_tools_and_prompts.py
|   |-- 0063_add_agent_parametrizer_prompt.py
|   |-- 0064_add_chat_agent_run_model.py
|   |-- 0065_add_chat_wrapped_agent_tools.py
|   |-- 0066_add_keyboard.py
|   |-- 0067_add_multi_turn_demo_prompts.py
|   |-- 0068_add_googler_tool.py
|   |-- 0069_add_googler_agent.py
|   |-- 0070_add_telematini.py
|   |-- 0071_acpx_skills.py
|   |-- 0072_add_acpx_demo_prompts.py
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```
-- 0073_acpx_demo_gemini_uplift.py
-- 0074_simplify_demo_prompts.py
-- 0075_add_acpx_tool_surface.py
-- 0076_add_acpxer.py
-- 0077_add_whatstlatamini.py
-- 0078_add_chat_agent_keyboard_tool.py
-- 0079_add_chat_agent_mouser_tool.py
-- 0080_add_chat_agent_sleeper_tool.py
-- 0081_add_window_present_and_run_wait_tools.py
-- 0082_add_chat_agent_j_decompiler_tool.py
-- 0083_add_de_compressor.py
-- 0084_add_chat_agent_de_compressor_tool.py
-- 0085_add_unrealer.py
-- 0086_add_chat_agent_unrealer_tool.py
-- 0087_add_unrealer_demo_prompt.py
-- 0088_add_reviewer.py
-- 0089_add_analyzer.py
-- 0090_add_reviewer_analyzer_demo_prompts.py
-- 0091_add_playwrighter.py
-- 0092_add_chat_agent_playwrighter_tool.py
-- 0093_add_windower.py
-- 0094_add_chat_agent_windower_tool.py
-- 0095_add_windower_playwrighter_demo_prompts.py
-- 0096_add_director_virtuoso_demo_prompts.py
-- 0097_add_kalier.py
-- 0098_add_chat_agent_kalier_tool.py
-- 0099_add_kalier_demo_prompts.py
-- 0100_add_unrealer_extended_demo_prompts.py
-- 0101_add_stm32er.py
-- 0102_add_chat_agent_stm32er_tool.py
-- 0103_add_stm32er_demo_prompts.py
-- 0104_fix_stm32er_hil_serial_proof.py
-- 0105_add_esp32er.py
-- 0106_add_chat_agent_esp32er_tool.py
-- 0107_add_esp32er_demo_prompts.py
-- __init__.py
-- monitors/
--   -- monitor_sql.py
-- opus_client/
--   -- claude_opus_client.py
--   -- examples.py
--   -- README.md
-- rag/
--   -- __init__.py
--   -- config.py
--   -- factory.py
--   -- interaction.py
--   -- interface.py
--   -- loaders.py
--   -- prompts.py
--   -- retrieval.py
--   -- splitters.py
--   -- utils.py
--   -- chains/
--     -- base.py
--     -- basic.py
--     -- history_aware.py
--     -- unified.py
-- services/
--   -- __init__.py
--   -- agent_contracts.py
--   -- agent_paths.py
--   -- answer_analyzer.py
--   -- filesystem.py
--   -- flow_compiler.py
--   -- flow_spec.py
--   -- response_parser.py
-- skills/
--   -- __init__.py
--   -- frontmatter.py
--   -- harness.py
--   -- io_contract.py
--   -- registry.py
-- skills_pkg/
--   -- _meta/
--     -- lint.py
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-- schema.json
-- acp_router/
--   |-- SKILL.md
-- code_review/
--   |-- SKILL.md
-- create_new_agent/
--   |-- SKILL.md
-- create_new_mcp/
--   |-- SKILL.md
-- github/
--   |-- SKILL.md
-- gmail/
--   |-- SKILL.md
-- hello_world/
--   |-- SKILL.md
-- jira/
--   |-- SKILL.md
-- kali_pentest/
--   |-- SKILL.md
-- notion/
--   |-- SKILL.md
-- security_audit/
--   |-- SKILL.md
-- setup_new_acpx_key/
--   |-- SKILL.md
-- skill_creator/
--   |-- SKILL.md
--   |-- scripts/
--     |-- quick_validate.py
-- slack/
--   |-- SKILL.md
-- summarize/
--   |-- SKILL.md
-- tlamatini_allowed_hosts_tighten/
--   |-- SKILL.md
-- tlamatini_csrf_exempt_audit/
--   |-- SKILL.md
-- tlamatini_exec_report_row_adder/
--   |-- SKILL.md
-- tlamatini_flow_from_objective/
--   |-- SKILL.md
-- tlamatini_flw_doctor/
--   |-- SKILL.md
-- tlamatini_new_acp_agent/
--   |-- SKILL.md
-- tlamatini_planner_trace_replay/
--   |-- SKILL.md
-- tlamatini_static_version_bumper/
--   |-- SKILL.md
-- todoist/
--   |-- SKILL.md
-- trello/
--   |-- SKILL.md
-- weather/
--   |-- SKILL.md
-- static/
--   |-- agent/
--     |-- css/
--       |-- agent_page.css
--       |-- agentic_control_panel.css
--       |-- login.css
--       |-- skills_dialog.css
--       |-- tools_dialog.css
--       |-- welcome.css
--     |-- img/
--       |-- spinner.svg
--       |-- Tlamatini.ico
--     |-- js/
--       |-- acp-agent-connectors.js
--       |-- acp-canvas-core.js
--       |-- acp-canvas-undo.js
--       |-- acp-control-buttons.js
--       |-- acp-file-io.js
--       |-- acp-flow-snapshot.js
--       |-- acp-globals.js
--       |-- acp-layout.js
```

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```

|-- acp-parametrizer-dialog.js
|-- acp-running-state.js
|-- acp-session.js
|-- acp-undo-manager.js
|-- acp-validate.js
|-- agent_page_canvas.js
|-- agent_page_chat.js
|-- agent_page_context.js
|-- agent_page_dialogs.js
|-- agent_page_init.js
|-- agent_page_layout.js
|-- agent_page_state.js
|-- agent_page_ui.js
|-- agentic_control_panel.js
|-- canvas_item_dialog.js
|-- chat_page_runtime_poller.js
|-- contextual_menus.js
|-- shared-runtime-dialogs.js
|-- skills_dialog.js
|-- tools_dialog.js
|-- sounds/
|   |-- hypervisor_alert.wav
|   |-- notification.wav
|-- video/
|   |-- XAIHT-Tlamatini.mp4
|-- telegram/
|   |-- AztecTlamatini.txt
|   |-- config.yaml
|   |-- telegram_receiver.py
|-- templates/
|   |-- agent/
|       |-- agent_page.html
|       |-- agentic_control_panel.html
|       |-- login.html
|       |-- welcome.html
|-- TlamatiniSourceCode/
|   |-- README.md
-- DB/
|   |-- Older/
|       |-- README.md
|   |-- ToLoad/
|       |-- README.md
-- jd-cli/
|   |-- jd-cli.bat
|   |-- jd-cli.jar
-- tlamatini/
|   |-- __init__.py
|   |-- asgi.py
|   |-- context_processors.py
|   |-- logging_filters.py
|   |-- middleware.py
|   |-- settings.py
|   |-- urls.py
|   |-- wsgi.py

```